

Jake DeMontigny

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Professional Summary

Highly motivated and results-driven Mechanical Designer with 5+ years of experience specializing in 3D modeling, BOM management, and technical drafting within the oil/gas, manufacturing, and mining sectors. Proven ability to translate complex field data and hand sketches into precise As-Built and fabrication-ready drawings. Expertly skilled in SolidWorks, AutoCAD, Inventor, and SAP, leveraging these tools to drive project efficiency, manage inventory, and ensure on-time delivery of high-quality mechanical designs. Seeking to leverage strong problem-solving and collaboration skills to excel in a challenging design role.

Professional Experience

Interface Testing Services, Sarnia ON

Inspection Drafter (Contract) (June 2025 - November 2025)

- Executed precise modification and creation of over 200 Isometric Piping Drawings for Imperial Oil Sarnia, ensuring adherence to client specifications.
- Successfully transitioned over 100 existing Isometric drawings to As-Built status by interpreting and incorporating field inspection team data.
- Generated over 100 new Isometric drawings directly from field team hand sketches, strictly following Imperial Oil drafting policies and block standards.

Plastruct Polyzone, Vineland ON

Mechanical Designer (August 2024 - June 2025)

- Created detailed 3D models of Tivar-88 Liners for various hoppers and trucks using SolidWorks, AutoCAD, and Inventor, ensuring optimal fit and function.
- Translated complex field measurements into precise 3D models and subsequent 2D shop drawings, reducing fabrication errors.
- Streamlined project workflow by managing the Bill of Materials (BOM) for each job and tracking hours to maintain project efficiency and control costs.
- Collaborated directly with production teams to resolve manufacturing challenges and provide accurate fabrication support, improving communication.
- Leveraged SAP software for inventory management and project tracking, enhancing material availability and scheduling.

Kernic Systems, Burlington ON

Mechanical Designer (April 2024 - July 2024)

- Produced 2D plant layouts and ductwork designs using AutoCAD with custom company blocks along with the Nordfab plug-in.
- Developed multiple detailed bills of materials for current and future projects using Microsoft Excel.
- Attended weekly project meetings to establish schedules and address field installer concerns.
- Designed complex 3D models of equipment and duct system layouts using Autodesk Inventor for client visualization and design validation.
- Developed and maintained multiple detailed Bills of Materials and costing sheets in Microsoft Excel, supporting accurate project quotations and bid submissions.

Lycopodium Minerals, Mississauga ON

Junior Mechanical Designer (January 2023 - April 2024)

- Created and modified over 100 PFD's and P&IDs; for mining projects, including updating drawings to "As-Built" status.
- Designed steel plate work and ROM Bins for gold processing plants using Open Buildings Designer.
- 3D Modeled vendor equipment using provided 2D drawings and produced 2D site setup drawings for multiple projects.
- Participated in design review meetings, learning and applying best practices for plant layouts.

MCNS Environmental, Tobermory ON

CAD Drafter (May 2018 - August 2021)

- Produced site layout drawings for industrial dust collection systems.
- Developed PFDs and P&IDs for steel projects in Nanticoke and Etobicoke Ontario.

Education

- Mohawk College, Hamilton ON
→ Mechanical Engineering Technology (January 2022 - December 2023)
- McMaster University, Hamilton ON
→ Automotive and Vehicle Engineering (September 2017 - December 2021)

Technical Skills

- Software: SolidWorks, AutoCAD, Inventor, Navisworks, Bentley MicroStation, SAP.
- Programming: C++, C, Java, Python, HTML, React.
- Other: Microsoft Office Suite, 3D Modelling, Technical Drawings, BOM Management.

Additional Skills

- Strong problem solving and time management abilities.
- Collaborative and independent work capabilities.
- Quick learner with a focus on efficiency and accuracy.
- Positive attitude with excellent organizational skills.